Parmida Davarmanesh

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MIT Ph.D. candidate in AI (LLM post-training, steering, interpretability) with first-author papers in PNAS and at a NeurIPS workshop; NSF Graduate Research Fellow and prize-winning speaker. Seeking full-time ML/AI research scientist roles. **Available Summer 2027.**

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

- Ph.D., Computer Science (LLM post-training, mechanistic interpretability, tokenization; advisor: Dr. Ashia Wilson) Expected 2027
- M.S., Computer Science (thesis: 3D genome organization in 4-cell mouse embryos; advisor: Dr. Caroline Uhler) 2023

University of Michigan, Ann Arbor, MI

- B.S., Honors Mathematics (minor in Computer Science) · GPA 3.96/4.0 2021

PUBLICATIONS

1. **Parmida Davarmanesh**, Ashia Wilson, and Adityanarayanan Radhakrishnan. "Efficient and Accurate Steering of Large Language Models Through Attention-Guided Feature Learning." *Proc. Natl. Acad. Sci. U.S.A.* 123(42): e2609978123 (2026).
2. **Parmida Davarmanesh**, et al. "Motif-Aware Tokenization of the Genome: Towards Interpretable Modeling of Gene Regulation." *NeurIPS 2025 2nd Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences* (2025).
3. **Parmida Davarmanesh**, Qian Lin, Irene Tenison, Gabriel Jabbour, and Ridwan Alam. "Detection of Acute Myocardial Infarction Using Deep Learning on Lead-I ECG Data." *IEEE BSN* (2024), poster presentation.
4. **Parmida Davarmanesh**, Gabriel Jabbour, Qian Lin, Irene Tenison, Collin Stultz, and Ridwan Alam. "Detection of Acute Myocardial Infarction and Ischemia From Lead-I ECG Using Deep Learning." *Circulation* 150(Suppl 1): Abstract 4125281 (2024).
5. Shengpu Tang, **Parmida Davarmanesh**, Yanmeng Song, Danai Koutra, Michael W. Sjoding, and Jenna Wiens. "Democratizing EHR Analyses with FIDDLE: A Flexible Data-Driven Preprocessing Pipeline for Structured Clinical Data." *JAMIA* 27(12): 1921–1934 (2020).

RESEARCH EXPERIENCE

Massachusetts Institute of Technology, Cambridge, MA 2025 – Present

Ph.D. Thesis Research: LLM Post-Training, Steering & Interpretability

Collaborator: Dr. Adityanarayanan Radhakrishnan

- Raised the percentage of successfully steered concepts from under 50% to ~95% (512 concepts, Llama-3.1-8B), nearly doubling the prior state of the art with a computationally efficient method that scales to 70B-parameter models (PNAS, 2026).
- Benchmarked the method against existing steering methods, outperforming them on coding and jailbreaking tasks.
- Currently post-training LLMs on mathematical reasoning using LoRA with GRPO, ORPO, and SFT objectives and prompt optimization, as part of thesis work on recursive self-improvement with reinforcement learning from verifiable rewards (RLVR).

MIT and the Broad Institute of Harvard and MIT, Cambridge, MA 2023 – 2025

Biologically Meaningful Tokenization of the Genome for More Efficient Training of Genomic LLMs

PIs: Dr. Bo Xia and Dr. Ashia Wilson

- Trained a transformer from scratch on the human reference genome (~1 TB) using distributed data parallelism on a Slurm cluster.
- ;Created a novel data-driven tokenization of the genome that uses known motif annotations to encode the genome as continuous-valued representations.
- Improved the training efficiency of genomic LLMs (FLOPs and model size) by shortening sequence length relative to other tokenization methods (e.g., one-hot, k-mer, BPE).
- Improved the interpretability of tokens on both linguistic and biological tasks and metrics.

Independent Project, MIT, Cambridge, MA Feb 2024 – 2025

Detection of Myocardial Infarction Using Deep Learning on Lead-I ECG Data

Collaborators: Dr. Collin Stultz, Dr. Ridwan Alam, and Dr. Gabriel Jabbour

- Performed signal processing and trained a ResNet on the novel task of detecting myocardial infarction (heart attack) and ischemia from lead-I ECG alone, a signal attainable from smartwatch devices.
- Established a benchmark with an AUC of 0.88 on the PTB-XL test set and 0.90 on an external dataset (PTB).
- Developed the project into a venture, which was a finalist and Audience Choice Award winner at the MIT $100K Accelerate Entrepreneurship Competition.

Thermo Fisher Scientific, Berlin, Germany June – Aug 2024

Macromolecule Classification and Orientation Estimation in Cryo-ET Tomograms

Mentor: Dr. Noushin Hajarolasvadi

- Worked with a team and trained 3D CNNs (Keras) on ~400,000 synthetic sub-tomograms: 0.99 per-class accuracy, 0.98 mean 6-DoF orientation correlation; interpreted models with GradCAM++ and LRP; visualized 3D molecules, isosurfaces, and heatmaps in Amira.
- Presented results to industry experts within and outside the domain.

Broad Summer Research Program, Broad Institute of Harvard and MIT June 2020 – April 2021

Blind Three-Dimensional Point Spread Function Estimation from Fluorescent Microscopy Image Stacks Using Generative Adversarial Networks

Mentor: Dr. Mehrtash Babadi

- Generated a synthetic dataset of 3D microscopy images; built and trained a novel framework in PyTorch that uses a generative adversarial network to estimate the point spread function non-parametrically; presented the work to an interdisciplinary team of scientists.

TEACHING

Teaching Assistant, Linear Algebra, MIT Fall 2026

- Teach recitations, hold office hours, and build visuals and simulations that make abstract concepts tangible for students.

AWARDS

First Prize, ITA Graduation Day Talks (Information Theory and Applications Workshop) 2026

Best Paper Award, MIT LIDS Student Conference 2026

Finalist and Audience Choice Award winner ($1,000), MIT $100K Accelerate Entrepreneurship Competition 2025

- Conducted customer discovery with cardiologists, patent attorneys, and angel investors on clinical utility and go-to-market strategy for an ECG-based heart-attack detection device.
- Served on the organizing committee (2024).

NSF Graduate Research Fellowship Program ($37,000 annually) 2023 – 2026

MIT EECS Distinguished Graduate Fellowship ($38,000) 2021

Outstanding Undergraduate Researcher Award (honorable mention), Computing Research Association 2020

Iranian Scholarship Foundation ($5,000 annually) 2018 – 2020

Maryam Mirzakhani Annual Memorial Scholarship ($2,500), Iranian American Scholarship Fund 2017 – 2018

TECHNICAL TALKS

INFORMS, San Francisco, CA Nov 2026

ML Tea Talk, MIT, Cambridge, MA Oct 2026

MIT Statistics and Data Science Conference, Cambridge, MA April 2026

Information Theory and Applications, San Diego, CA Feb 2026

LIDS Student Conference, MIT, Cambridge, MA Jan 2026

LEADERSHIP

MIT Iranian Association May 2026 – Present

Founder and Treasurer

- Founded the association and manage its $4,500 budget; organized 4 events (~50 attendees typically), with 3 more planned.

MIT Volunteer Consulting Group, Cambridge, MA

Project Leader *Sept 2023 – Feb 2024*

- Framed the indication-expansion case with an early-stage VC-backed biotech company (valued at $3.7M); recruited and led a team of 6 consultants during a 6-week engagement to produce a strategy document, “MIT VCG Report.”

Consultant (top 13% of applicants) *May – June 2023*

- Collaborated with a team of 6 consultants to shape customer relationship strategy for a science education non-profit via market research and customer segmentation analysis; conducted interviews with 20+ key opinion leaders.

Electrical Engineering and Computer Science Graduate Student Association, MIT 2022 – 2023

Social Chair (2022), Publicity Chair (2023)

- Organized monthly social events for community building and networking, attended by 40+ graduate students on average.

SKILLS

Technical: Python, PyTorch, Git · deep learning, fine-tuning (LoRA, SFT) · RL (GRPO, ORPO, RLVR) · LLM steering, interpretability